Deep CNN-LSTM Network for Indoor Location Estimation Using Analog Signals of Passive Infrared Sensors

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***Abstract*—Indoor localization is a crucial component of IoT applications in many areas, such as healthcare, energy management, and security control. Passive infrared (PIR) sensor has been employed for a location estimation due to its cost effectiveness, low power consumption, and low electromagnetic interference. Compared with its binary output, PIR analog output which is an output voltage generated by a PIR sensor when its sensing elements detect changes in temperature in an environment can provide more information regarding a person's location. However, only a few works focus on using analog signals for location estimation. During the past several years, deep learning approaches have emerged and achieved outstanding results in many applications. In this article, we harness the power of deep learning and propose a deep CNN-LSTM architecture for PIR-based indoor location estimation. In our architecture, an upper CNN network can extract features from PIR analog output automatically while a lower LSTM network can learn temporal dependencies between the extracted features. To evaluate the feasibility and performance of our proposed method, we conduct four different sets of experiments. Our results show that the proposed method can efficiently handle complex cases and can achieve the mean distance error of 0.23 m, and 80% of distance errors are within 0.4 m.**

***Index Terms*—Device-free, indoor, localization, location estimation, passive infrared (PIR).**

I. Introduction

THE FAST improvement of IoT devices has empowered many useful indoor applications in areas, such as healthcare, energy management, and security control [1], [2]. To support such applications, accurate person's whereabouts are also crucial and need to be reported so that information such as medical alerts can be analyzed and handled immediately by interested parties [3], [4]. Various surveillance technologies have been employed to enable location estimation in an indoor environment. Wearable or mobile devices, such as smartwatches and mobile phones, allow us to estimate the location of a person by determining the current location of the device. However, it is not always user-friendly for a person to carry a wearable or mobile device around a premise because it is not comfortable, and there is a risk of device loss [5]. On the contrary, these drawbacks can be overcome using radio frequency (RF), WiFi, and sensor devices, such as RFID tags, surveillance cameras, passive infrared (PIR) sensors, and electric field sensors. When privacy is concerned in certain indoor areas, passive sensing devices and sensors, such as PIR and WiFi, are commonly adopted because they sense the presence of a person without recording nor revealing their identities or appearance.

Major advantages of PIR sensors include a fast response time, low power usage, and strong tolerance against electromagnetic interference [6]. Despite these advantages, PIR sensors have relatively low to moderate coverage distance, and their detection accuracy can be affected by a rise and drop in temperature in a surrounding environment. PIR sensors have been employed successfully to locate or track the location of a person. However, most existing work only focuses on the digital outputs of PIR sensors due to their low computational cost. A person's location can be either estimated by analyzing overlapped areas of the activated PIR sensors or located in the proximity of the activated sensors, however, a sophisticated sensors node or dense sensors deployment is needed [7]–[10]. As a result, their complicated installation makes it impractical in a real-world setting. On the other hand, as characteristics of PIR analog outputs are influenced by factors, such as the distance between a person and a PIR sensor, walking speed, and walking direction [11], it is possible to leverage features of the analog outputs to estimate a person's location. Commonly used filtering algorithms, such as Kalman filtering and particle filtering (PF) can be employed to estimate a location based on characteristics of PIR analog outputs. However, there is not sufficient information on the specifications of commercial off-the-shelf (COTS) PIR sensors to formulate an observation model that describes a relationship between a distance and a PIR analog output [9]. To obtain a specification of a PIR sensor, we require an extensive experiment and simulation. In traditional machine learning methods, one of the problems is to find out the appropriate features for each application which requires us to extract and test features manually. As a result, it is labor intensive and time consuming [12]. In addition, analog signals are quite ambiguous. To obtain accurate location estimation, we need to know about the dependency between locations and raw PIR outputs in each time step. However,

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traditional machine learning algorithms cannot capture this complex dependency very well [13]. Thus, they produce inaccurate estimations. Therefore, it is quite challenging to use analog outputs for location estimation. To the best of our knowledge, only a few works focus on the analog outputs of PIR sensors [14], [15].

During the past several years, various deep learning techniques have been developed and achieved promising results to address the localization problem based on RF signals [12], [13], [16]. Compared with traditional machine learning approaches, deep learning methods can automatically extract discriminative features from raw data without the intervention of domain experts and can learn complex data representations through hierarchical multilayers even though there are small training samples (e.g., less than 1000 samples of data [17], [18]) and various characteristics of data [19]. As a result, the performance of the deep learning methods for localization problems generally outperforms traditional machine learning methods [12], [20]. However, some machine learning methods such as support vector regression (SVR) may achieve better performance due to small data sets, less complex problems, or requiring extensive feature engineering [20], [21].

In this article, we aim to leverage deep learning to address indoor location estimation based on analog signals of PIRs. We propose a two-phase framework for indoor location estimation which consists of offline and online phases. In the offline phase, raw PIR outputs are preprocessed and used as a training data set. The offline phase consists of our CNN-LSTM architecture, which can process the raw outputs directly to estimate the location of a person. To avoid the need for manual feature extraction, we design the upper convolutional network (CNN) level to automatically extract discriminative features from raw PIR outputs. Then, we incorporate a long short-term memory (LSTM) network in the lower level of the proposed architecture to capture the sequential correlation of the extracted features from the upper layer. As a result, the dependencies between locations and raw PIR outputs can be modeled appropriately. In the online phase, we adopt a threshold method to detect a person's movement. However, PIR signals are quite noisy and contain nonsymmetrical positive and negative phases. Thus, it is hard to set a threshold value properly. To address this issue, we use instantaneous energy instead of raw PIR outputs to simplify threshold value setting and reduce false detection. Once the CNN-LSTM model has been trained to estimate the location of a person, the median absolute deviation (MAD) method [22] is employed to detect and filter out infeasible estimated locations to enhance the estimation accuracy.

Our main contributions are summarized as follows.

1) We present a two-phase framework for indoor location estimation using PIR analog outputs.
2) We propose a deep CNN-LSTM architecture that can extract discriminative features from PIR analog outputs and learn temporal patterns of the extracted features to estimate a person's location.
3) We conduct a comprehensive set of experiments mimicking possible real-world scenarios to evaluate the feasibility and effectiveness of our proposed approach.

The remainder of this article is organized as follows. Section II discusses the related works. Section III describes the problem formulation. The proposed methodology is presented in Section IV. The experiments and their results are reported in Sections V and VI. Finally, Section VII provides some concluding remarks.

II. Related Works

To realize indoor location estimation, different technologies are employed, such as RF, surveillance cameras, and infrared sensors. In this section, we focus our discussion only on the classes of location estimation techniques that can be applied to the PIR technology.

Proximity techniques are the simplest location estimation method. We require a dense sensor deployment in the monitored area where the position of each sensor is known. A threshold method is mainly used to identify affected sensors caused by a person standing or moving nearby sensors. Then, the location of a person can be estimated to be nearby or at the position of an activated sensor. This type of technique can be used with RFID [23], [24], ZigBee/IEEE 802.15.4 [25], and PIR [8], [26]. Due to a dense sensor deployment, it is rather impractical for deployment in a large area.

Angle of Arrival (AoA)-based techniques measure angles of arrival of each sensor node, and lines can be drawn w.r.t. the measured angles. Then, the location of a person can be determined from the intersection of these lines. Wang *et al.* [27] exploited a multipath problem of RFID tags and extracted AoA using an MUSIC algorithm. Li *et al.* [28] measured the AoA of a Wi-Fi signal using the Dynamic Music algorithm. Adib and Katabi [29] employed smoothed music algorithm to extract the AoA of Wi-Fi signals reflected from a moving person to localize a person through walls. Some PIR works utilize binary PIR outputs to measure an angle of detection. Hao *et al.* [7] developed two types of infrared modules consisting of four PIR sensors in two columns. Type I module, which has a Field of View (FoV) of each sensor separated by 16 degrees, is specialized for locating multiple persons. Type II module, which has an FoV of each sensor covered by pseudo-random coded masks, is designed for target identification. They employed a Bayesian tracking framework to track each person while a hidden Markov model (HMM) was used to identify a person. Yang *et al.* [30] created 12 detection zones around a sensor node by overlapping the FoV of six PIR sensors. The detection zones are used to determine the detection angle, and a person's location can be determined based on the intersections of these angles. However, their method may produce false intersections. To overcome this drawback, Yang *et al.* [32] employed a naïve Bayes classifier [31] or artificial neural networks to scope down estimated locations to possible subareas.

Model-based techniques can estimate a person's location using mathematical models that describe relationships between the positions of a person and sensor outputs. For the RF technology, two types of RF phenomena, including the shadowing effect and diffraction are employed to formulate mathematical models. Shadowing models attempt to describe RF attenuation

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caused by a person obstructing RF links. There are various variations of shadowing models, such as the excess path delay model [33], [34], elliptical model [35], [36], exponential-Rayleigh model (ERM) [37], Saddle Surface model (SaS) [38] and imaging [39]. Diffraction models aim to characterize RF signals that are diffracted by a side of a human body, and a few works focused on models based on the diffraction effect [40], [41]. Recent work in the PIR technology also adopted a model-based approach. The PIRATES localization system was proposed in [15]. This localization system utilized a new azimuth change measurement which is the absolute difference in the azimuth of a person with respect to a PIR sensor at two locations. A mathematical model that describes a relationship between azimuth change and a person's location was presented, and location estimation is performed using a particle filter technique.

Fingerprinting methods are commonly used techniques for the RF technology, especially Wi-Fi. It was proposed by [42] to locate a person who carries an RF device. Later, Moussa and Youssef [43] leveraged the received signal strength (RSS) measurement of existing Wi-Fi infrastructure and were the first to adopt a fingerprinting method in a device-free manner. Based on this initial work, Youssef and Agrawala [44] introduced a Horus system that uses a center of mass technique with a time-averaging window to improve the localization's accuracy. Then, the fingerprint method has been further improved in [45] to support a large-scale area. Nonlinearity of the impact subjects on RSS is addressed in [46] and [47] to minimize time and efforts for fingerprint collection. Channel state information (CSI) was initially investigated in [48] to locate a person with a mobile Wi-Fi device because it is more stable and accurate compared with RSS. Using CSI amplitudes, a probabilistic method was used in [49] and [50] to map the fingerprints, while support vector machine (SVM) was employed in [21]. For PIRs, some works used techniques similar to the fingerprint methods. PIR analog outputs were initially explored in [51] to estimate the distance of a person that walks past two PIR sensors attached to opposite walls of a hallway. Distances between the opposite sensors are grouped into three classes, including close-to-sensor 1, middle, and close-to-sensor 2. Then, they employed three classifiers, including Naive Bayes, SVM, and k-nearest neighbor (k-NN), to classify the distances. The characteristics of PIR analog outputs were investigated extensively in [11]. This work found that the amplitude, phase, and frequency of PIR analog outputs can be affected by the distance, speed, and direction of a moving person. Some localization experiments are performed to validate the findings in this work using the same classifiers as in [51].

Recent advances in deep learning have shown several advantages for location estimation, such as improved accuracy and automatic feature extraction. An autoencoder network was employed in [12], [16], and [52] using CSI amplitudes, CSI phase, and RSS, while a deep CNN was adopted in [53] to estimate a location from CSI images. Chen *et al.* [13] extracted sequential local features using a sliding window and employed a deep LSTM to learn temporal dependency between the extracted features. ResNet was used with signal-to-noise ratios (SNRs) measurements to estimate a location as a label or coordinate. All of the above works reported an improvement in accuracy of around 20%–30% compared with traditional machine learning methods. To the best of our knowledge, the use of a deep learning approach for PIR-based location estimation is still unexplored.

III. Problem Formulation

Heat emanated from a moving person can be captured by a PIR sensor. Then, its analog outputs are generated with respect to the orientations of the PIR analog sensor and the distance between the person and the sensor [11]. Thus, we can arrange each PIR sensor to monitor each part of a monitored area and analyze the raw signals from each sensor to estimate a person's location. Based on this finding, we are motivated to locate a moving person using a sensor node that consists of multiple PIR sensors. The sensor node can be attached to a ceiling above the monitored area to avoid its line of sight being obstructed by furniture.

A monitored area A can be divided into a set of grids $G = \{g_i | i = 1, \ldots, n\}$ with a center coordinate at $g_i = (x_i, y_i)$. A grid i is further divided into a set of cells $C = \{c_j | j = 1, \ldots, 4\}$ with a center coordinate at $c_j = (x_j, y_j)$ where each cell has a size of 50 cm $\times$ 50 cm which fit the average size (standing) person. A PIR node is located at the center of the monitored area, and the node consists of m sensors. When a person walks across the monitored area, it generates a set of reference measurements $S_{\text{ref}} = \{s_k | k = 1, \ldots, m\}$ where s_k is a raw signal vector from a PIR sensor k. The raw signal vector can be represented as $s_k = [s_{k1}, s_{k2}, s_{k3}, \ldots, s_{kt}]$ where t is a time step. Given a set of current measurements S_{current}, we want to estimate the location of a person for each time step.

IV. Methodology

In this section, we discuss the methodology of our deep learning-based PIR location estimation. First, we introduce a location estimation process in which we collect training data to train a model in an offline phase, and a location of a person is estimated in an online phase. Second, we present our proposed CNN-LSTM architecture in the offline phase. Finally, we discuss the underlying techniques that are employed to support the online phase.

A. Proposed PIR Location Estimation Process

Our deep learning-based PIR location estimation process consists of two phases as shown in Fig. 1: 1) the offline phase and 2) the online phase.

In the offline phase, a person walks in a monitored area according to predefined instructions. Raw output signals are generated and collected from a PIR sensor node. Then, the raw outputs are preprocessed into a suitable training data structure. Next, our CNN-LSTM network is trained using the training data set.

In the online phase, the PIR node generates and sends the raw output signals, with or without a person walking in the monitored area, to a sink node periodically. When we receive

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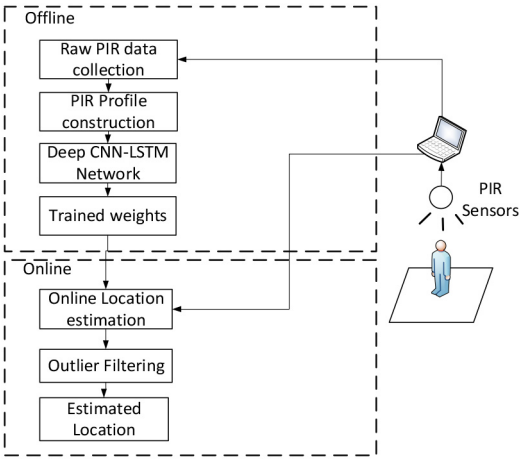


Fig. 1. Overview of PIR-based location estimation.

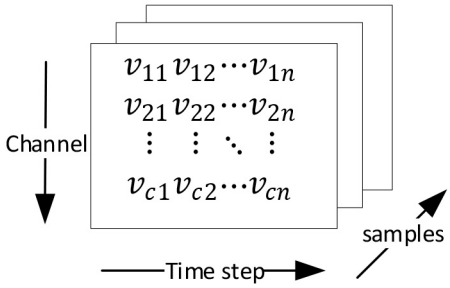


Fig. 2. Matrix of training data.

the raw outputs, the movement of a person can be detected by a threshold technique, which was successfully applied in several papers [25], [51]. In general, we can directly specify a threshold value to detect the changes in raw signals in a threshold method. However, the signals can contain both positive and negative pulses, and the amplitudes of these pulses are varied. Thus, it is difficult to set a proper threshold value. Instead, we apply Hilbert–Huang transform (HHT) [54] on the signals to obtain the Hilbert spectrum that captures characteristics of the signals, such as the amplitude and the instantaneous frequency as a function of time. As a result, we can extract the instantaneous energy of the signals over time which can be used to indicate a movement of a person. Once a movement is detected, the signals are fed into the trained model to predict estimated locations as 2-D coordinates. Then, we filter outliers to improve the accuracy of the location estimation.

B. Offline Phase

In the offline phase, we focus on collecting raw PIR signals for training our deep learning model to estimate the location of a person in a monitored area. First, we describe the process of a PIR profile construction. Second, we discuss the details of our proposed deep learning architecture. Third, a model training process is explained.

1) PIR Profile Construction: To construct a training data set, a person walks across a monitor area in different directions, and raw PIR signals are collected. We assume that a person's walking speed is constant to reduce the impact of speed on the PIR signals that is not our main focus to address in this work. For each walking direction, samples of PIR signals are stored in the following format, as shown below, where m represents an index number of each PIR sensor in a sensor node and n represents a time step:

$$V_{\text{Raw}} = \begin{Bmatrix} v_{11} & v_{21} & \cdots & v_{m1} \\ v_{12} & v_{22} & \cdots & v_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ v_{1n} & v_{2n} & \cdots & v_{mn} \end{Bmatrix}. \quad (1)$$

Next, the samples are cropped and aligned using a cross-correlation method to ensure that the samples of each direction are aligned as much as possible. A better alignment of the samples should help deep learning models to learn the correct features of the signal generated by a person moving in a certain location. Once the alignment is completed, we then form a training data matrix M with a size of $c \times n \times b$ where we treat sensors in a sensor node as c channel, n represents a time step of the PIR signals in each channel, and b is the number of samples, as shown in Fig. 2. In our case, the origin of the raw signals does not start at 0 but at 2.05 V. Unscaled inputs can cause unstable learning during a training process, and our model learns inappropriate weights, which can affect the accuracy of our deep learning model. Therefore, raw PIR output signals are rescaled to the origin of zero. However, we do not need to normalize the raw signal data because there is no difference in the scaling of readings from PIR sensors. In addition, a zero-padding is applied to the PIR profile because different walking directions result in different lengths of the profile signals. We also need to ensure the lengths of the profile data are equal because an inconsistent length of the data can affect the performance of our deep learning model [55]. Finally, we associate the center coordinates of reference locations to each time step to serve as a ground truth.

2) Proposed Network Architecture: To estimate a person's location using PIR analog signals, we adopt a deep learning-based approach based on a CNN and LSTM network. Our proposed architecture consists of two levels as shown in Fig. 3.

The upper part is for feature extraction via a CNN containing multiple convolutional layers, whose inputs are provided by the PIR node. There are three layers with the filter numbers 128, 256, and 512. Each convolutional layer gradually extracts dominant features which can be utilized to estimate a person's whereabouts. The lower part is a single LSTM layer that takes features extracted by the upper CNN. The LSTM layer focuses on learning temporal patterns of the extracted features to help differentiate a person's location at different time steps. We also use a dropout layer to reduce overfitting. Finally, a fully connected layer is employed to output the estimated coordinates for each time step. The details of each part are discussed in the following sections.

3) CNN Model: A CNN model consists of multiple convolutional layers. The core operation so-called "convolution" is performed in each convolution in the CNN model. The convolution is a linear operation that involves the dot product multiplication of a filter or a kernel that has its size less than the input data and is a set of weights with the input data.

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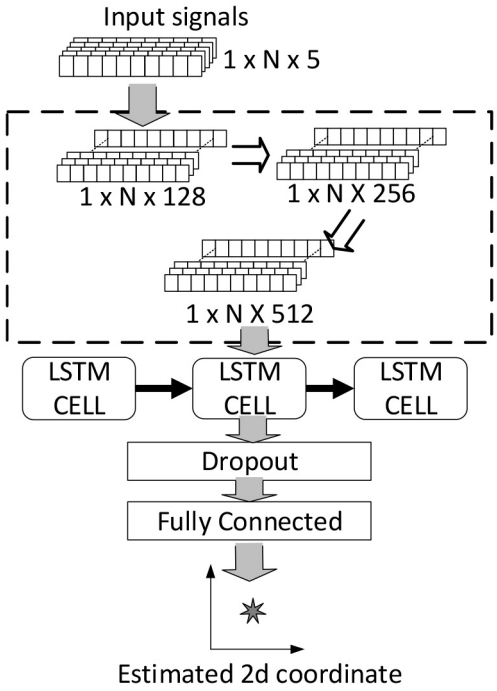


Fig. 3. Proposed deep CNN-LSTM architecture.

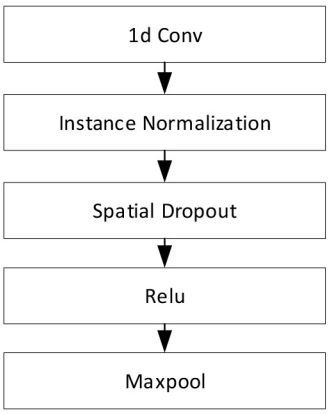


Fig. 4. Details of each CNN layer.

During the convolutional operation, the filter is slid systematically across the input to perform a multiplication at different parts of the input. As a result, this systematic operation allows the filter to discover some features in the input data.

In our model, for each convolutional layer, a 1-D convolution operation is performed along a time axis to extract features from the sequences of observation, as shown in Fig. 4. The size of the filter is set to 3, and the stride length is set to 1. We extract the feature maps according to the number of filters. A zero padding is applied to maintain the length of the input sequence. Then, instance normalization [56] is applied to the feature maps of 1-D convolution. For each observation, the mean u and variance σ^2 are calculated, and the input is normalized across each channel using the following formula:

$$\hat{X} = \frac{X - u}{\sqrt{\sigma^2 + \epsilon}} \tag{2}$$

where ϵ is added in the denominator for numerical stability and is an arbitrarily small constant. Compared with batch normalization, instance normalization can improve the performance (such as training time and convergence) of a deep learning model for time-series data [56]. After that, a dropout operation is performed on the normalized feature maps. This operation randomly ignores or drops some layer outputs to prevent overfitting and improve generalization error. Similar to [57], we consider that time-series data are highly correlated, and a normal dropout can lead to slow training time but does not prevent overfitting. In our model, we adopt a spatial dropout that extends a dropout operation across the entire feature map. Next, the feature maps are passed to the rectified linear activation (ReLU) function which is the most widely used activation function in a deep learning approach. The function returns the same value if it receives a positive value, otherwise, it returns zero if the value of an input is negative. It can be formulated as follows:

$$f(x) = \max(0, x). \tag{3}$$

There are three benefits for using the ReLU function [58]. First, it improves computational efficiency. Second, it prevents the vanish gradient problems. Finally, the advantages of sparsity can be exploited.

After that, down-sampling or pooling is performed to reduce the resolution of the feature maps while important features of the maps are maintained through translational and rotational invariants. As a result, pooling operation can improve the computational cost due to the reduced number of parameters and minimize the effect of overfitting. A 1-D max pooling with a window size of 2 and a stride length of 1 is performed across each feature map. Max pooling divides the feature map into subregions, and the maximum value of each region is calculated. Similar to the 1-D convolution, the zero-padding method is also applied for the same reason.

4) LSTM Network: LSTM is one type of recurrent neural network that specializes in learning temporal patterns from time series or sequential data. Similar to a traditional recurrent neural network, LSTM processes inputs and retains its memory of the received information. Then, it makes predictions according to information in its memory. LSTM has a mechanism that allows it to determine which information needs to be maintained or deleted. As a result, it is more superior to a traditional RNN in handling long sequences. A basic structure of vanilla LSTM is shown in Fig. 5. At a time step t, an input is denoted as x_t and h_t is a hidden state. Weights are denoted as w_f, w_i, w_C, and w_o. Biases are represented as b_f, b_i, b_C, and b_o. Once an input is received at time t, the previous time step information in the cell state at a time step $t-1$, denoted as C_{t-1}, is evaluated by a forget gate which can be expressed as follows:

$$f_t = \sigma\left(w_f\left[h_{t-1}, x_t\right] + b_f\right). \tag{4}$$

The purpose of the forget gate is to determine which information should be deleted from a cell state C_{t-1}. It returns either 1 or 0 as its output. If $f_t = 1$, the previous time step information is kept. Otherwise, the information is removed when $f_t = 0$. Next, two prerequisite tasks are executed before

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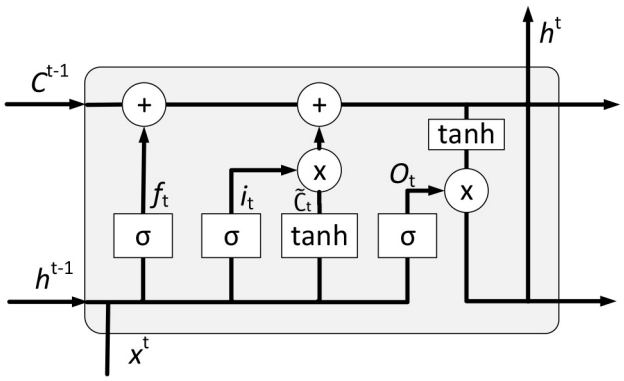


Fig. 5. Basic LSTM structure.

the cell state is updated. In the first task, an input gate determines which value needs to be updated and can be formulated as follows:

$$i_t = \sigma\left(w_i\left[h_{t-1}, x_t\right] + b_i\right) \tag{5}$$

where $\sigma(\cdot)$ is a sigmoid function. A candidate state is generated using the following equation:

$$\tilde{C}_t = \tanh\left(w_C\left[h_{t-1}, x_t\right] + b_C\right) \tag{6}$$

where $\tanh(\cdot)$ represents a tangent function. Then, the current state C_t is updated using the following equation:

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t \tag{7}$$

a multiplication of f_t and C_{t-1} may result in deleting some information in a cell state C_{t-1} while the multiplication of i_t and $\tilde{C}_t$ determine which information is important to maintain from $\tilde{C}_t$. Therefore, the current state can be updated by the addition of outputs of these two multiplications. To calculate the final hidden unit output h_t, we need both the current state C_t and the output of a sigmoid function o_t, which indicates what information the hidden state should keep, and its equation can be expressed as follows:

$$o_t = \sigma\left(w_o\left[h_{t-1}, x_t\right] + b_o\right). \tag{8}$$

Finally, the current hidden state h_t is computed using the following formula:

$$h_t = o_t \tanh(C_t). \tag{9}$$

The current cell state C_t and the current hidden h_t are then carried over to the next time step. By utilizing LSTM in our architecture, extracted PIR signal features can be stored in memory cells systematically while temporal dependencies between the features are also preserved. This memory cell concept is very crucial for PIR-based location estimation because the information of the past time steps can help improve the location estimation [13]. In our research, the sequences of the features extracted by the upper CNN level are fed into a single LSTM network, as shown in Fig. 3. We do not stack one LSTM layer on the other LSTM layer because it performs worse than a single layer due to the effect of overfitting [59]. Our LSTM uses a multiple-input multiple-output structure (MIMO) in which it processes the input sequences sequentially and returns the output sequences of the same length as its input. Finally, the outputs of the LSTM layer are passed to the fully connected layer to estimate the location for each time step.

5) Model Training: To train our CNN-LSTM, network parameters need to be optimized by minimizing a loss function. Specifically, the distance error between the estimated location and the actual location needs to be minimized to ensure the performance of a trained model. The adaptive moment estimation (Adam) optimizer [60] is employed for this task. This optimizer can be viewed as a combination of root mean square propagation (RMSprop) and stochastic gradient descent (SGD) with momentum because the learning rate is scaled using the squared gradients like RMSprop, and it takes advantage of momentum by using the moving average of the gradient instead of the gradient itself like SGD with momentum. The Adam optimizer is widely used in several works due to its excellent convergence and computational efficiency for training a deep neural network [13], [55], [59]. Given that θ_t is the parameter to be optimized and the gradient is denoted as g_t, the optimizer updates the parameter θ_{t+1} as described in the following steps:

$$\begin{aligned} m_t &= \beta_1 \cdot m_{t-1} + (1-\beta_1)\cdot g_t \\ v_t &= \beta_2 \cdot v_{t-1} + (1-\beta_2)\cdot g_t^2 \\ \hat{m}_t &= m_t / \left(1-\beta_1^t\right) \\ \hat{v}_t &= v_t / \left(1-\beta_2^t\right) \\ \theta_{t+1} &= \theta_t - \alpha \cdot \hat{m}_t / \left(\sqrt{\hat{v}_t} + \epsilon\right) \end{aligned} \tag{10}$$

where m_t and v_t are estimates of the first moment and the second moment of the gradient, $\hat{m}_t$ and $\hat{v}_t$ represent the corrected estimates, the decay factors are denoted as β_1 and β_2, respectively, α is a learning rate, and a small scalar denoted as ϵ is used to prevent division by zero. Mean square error is selected as our loss function to be minimized by the Adam optimizer. It can be expressed as follows:

$$\text{MSE} = \frac{\sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2}{n} \tag{11}$$

where n represents the number of samples, y_i is an actual coordinate, and $\hat{y}_i$ indicates an estimated coordinate.

C. Online Phase

We use the trained CNN-LSTM network to estimate a person's location in the online phase. In this section, we discuss the process of the online phase. First, we explain our technique of movement detection, followed by the technique for location estimation.

1) Movement Detection: For PIR-based location estimation, a person's movement affects the temperature of the surrounding environment. PIR sensors can sense this change and generate output signals. By analyzing the signals, we can leverage its features for location estimation. However, PIR signals are sensitive to a small temperature change, thus causing false detection and estimation even though there are no movements. Therefore, movement detection is the first crucial step in PIR-based location estimation. A threshold method is

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commonly used to determine each signal event to be either active or nonactive using a threshold value [25], [51]. PIR signals have both positive and negative phases. Based on our observation, the amplitudes of both phases are not always symmetrical. In addition, multiple PIR sensors may have different sensitivities. As a result, it is difficult to set the best suitable threshold values for each sensor.

To find the most appropriate threshold value, we rely on using instantaneous energy to provide a simple viewpoint to support motion detection. We observe that the instantaneous energy rises and falls according to a movement of a person, and its value is always positive. As a result, the energy is more suitable than voltage to indicate a movement of a person. The HHT technique [54], which is used to perform time-frequency analysis of a nonstationary signal, is adopted to measure instantaneous energy. To the best of our knowledge, this is the only suitable technique that can calculate the energy of signals. For each sensor, its signal is decomposed, via an iterative sifting process, into a finite number of intrinsic mode functions (IMF) by the empirical mode decomposition [61] as follows:

$$X(t) = \sum_{i=1}^{N} \mathrm{IMF}_i(t) + r_n(t) \tag{12}$$

where $X(t)$ is an original signal, $IMF_i(t)$ is the ith IMF component, and $r_n(t)$ is a residual. Then, the Hilbert transform (HT) that is used to determine the properties of a vibration system such as amplitude, instantaneous phase, and time-varying frequency is applied to each IMF component. In the HT, the analytical signal $Z(t)$ is computed as follows:

$$Y(t) = H[X(t)] = \int_{-\infty}^{\infty} \frac{x(u)}{\pi(t-u)} du \tag{13}$$

$$Z(t) = X(t) + iY(t) = A(t)e^{i\theta(t)} \tag{14}$$

where $H[X]$ is the Hilbert operator, the imaginary unit is denoted as i, and the amplitude and the instantaneous phase angle are represented as $A(t)$ and $\theta(t)$, respectively. Based on $Z(t)$ and $Z(t)$, the instantaneous energy and the instantaneous frequency can be expressed as follows:

$$E(t) = |A(t)|^2 \tag{15}$$

$$\omega(t) = \frac{\mathrm{d}\theta(t)}{dt}. \tag{16}$$

For each sensor, the instantaneous energy is calculated by averaging the energy of all IMF components. Then, we calculate the instantaneous energy across all sensors to provide a single energy value that can help define a threshold value by calculating the mean energy

$$\overline{E} = \frac{\sum_{i=1}^{n} E_i}{n} \tag{17}$$

where n is the total number of sensors, E_i is the instantaneous energy of the ith sensor. Once we have the mean energy across all the sensors, it is possible to determine a single threshold value for all the sensors as follows:

$$P(S, e) = \begin{cases} 1, & \text{if } \overline{E} > T \\ 0, & \text{otherwise} \end{cases} \tag{18}$$

where a threshold value is T, an event is denoted as e, S is a set of sensors, and $\overline{E}$ is the calculated mean energy.

2) Location Estimation: Once a movement is detected, the PIR signals are passed to the trained CNN-LSTM model to estimate the location of a person. Unlike the training phase, the input data is not the entire sequence generated by a person walking across the monitored area, i.e., the input data is a subset of the profile sequence. The length of the input data depends on the frequency of the location estimation, e.g., 0.5 s. In the first location estimation, a hidden state and cell state are initialized. Then, the CNN-LSTM model processes the input data to estimate the location as well as update both the hidden state and the cell state. The output of the model is a sequence of the estimated location that has the same length as its input. Each value in the output sequence is represented as a 2-D or x–y coordinate to indicate the location of a person at each time step, as shown in the following:

$$O = \begin{bmatrix} x_1 & x_2\ x_3 & \cdots & x_n \\ y_1 & y_2\ y_3 & \cdots & y_n \end{bmatrix}. \tag{19}$$

The updated hidden state and cell state of the LSTM layer are returned to be used by the next estimation, and the median value of the output sequence is stored. Then, we determine the next location estimation using the new input data, the updated hidden state, and the updated cell state. However, the hidden and cell states are updated based on the input data of the current time step only, but a relationship between the input data in consecutive time steps is not captured. Thus, the next estimated location may be less accurate if the input data of the next time step is ambiguous or does not have enough information. To address this issue, we update the hidden and cell states using two consecutive time steps, i.e., the current time step and the previous time step. As a result, we can obtain better location estimations compared with when we use only the current time step to update states of the LSTM layer. We found that two consecutive time steps yield the best results amongst longer time steps. When the number of time steps increases beyond three-time steps, the accuracy of location estimation starts to decline due to too much irrelevant information updated to hidden and cell states. When a person turns his body to change his walking direction or stops moving for a short moment, we observe that a person's movement may not be detected. If the person is detected in the previous time step and his recent position is known, we can consider that the person's location remains unchanged, and the stored median value from the previous estimation is assigned as the estimated location.

From our initial experimental results, some estimated locations are not feasible for a person to travel from the previous location, and these errors can affect the overall accuracy of location estimation. To improve the accuracy, infeasible locations need to be filtered out. However, we cannot filter the infeasible locations using a boundary of a walking distance because we do not have explicit knowledge of a person walking speed to determine the possible walking distance. To detect outliers, the method of the mean and standard deviation is commonly used, but the mean and standard deviation are strongly affected by outliers [22]. Thus, they are not a good

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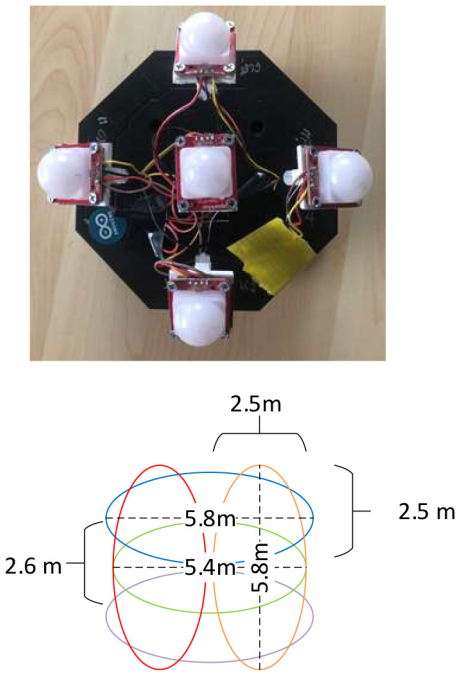


Fig. 6. PIR sensor node and its FoV.

indicator for outlier detection. Therefore, we apply a statistical method to detect outliers based on a scaled MAD that is insensitive to outliers, and MAD can be calculated as follows:

$$\text{Scaled MAD} = c * \text{MAD} \tag{20}$$

where MAD can be defined as median($|x_i - \text{median}(x)|$) and c is calculated as $-1/(\sqrt{2} \times (\text{erfc}^{-1}(3/2)))$. The estimated locations that exceed three-scaled MAD are identified as outliers. Then, we replace identified outliers with the median value.

V. Experiment Setup

In this section, we first describe our hardware and environmental setups. Then, we present our data collection process and scenarios used to evaluate our model.

A. Hardware and Environmental Setup

We develop a sensor node that consists of five PIR sensors (Model SEN-13968) from SparkFun and each sensor can generate the maximum voltage output of 5 V. We arrange the five sensors on a sensor node and their (overlapped) FoVs are projected to cover the monitored area, as shown in Fig. 6. Our designed FoVs create overlapping detection zones that can help in the generation of distinct signal patterns when a person moves in different zones. The FoVs are aligned symmetrically as much as possible to ensure that we have consistent patterns. We use Arduino MEGA 2560 microcontroller to synchronize the transmission cycle of the five sensors and combine the most recent measurements received from these sensors. The information from the microcontroller is transmitted to a desktop computer, which acts as a sink node that stores all voltages measurements for processing, via serial communication. The sampling rate is set to 60 samples per second because we want to ensure that we do not miss any important features in PIR raw signals, and this sampling rate facilitates a time synchronization for data labeling.

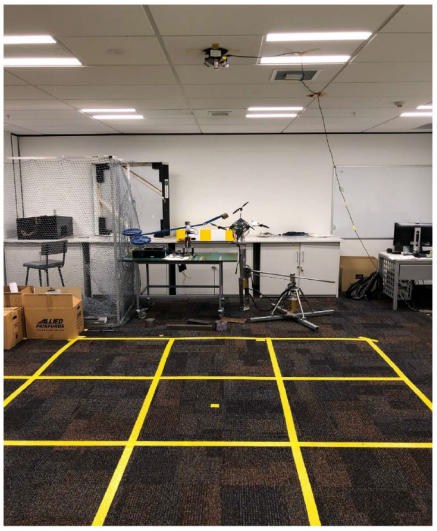


Fig. 7. Area monitored by a PIR node on the ceiling.

In our setup, we attach our PIR node on the ceiling with a distance of 2.5-m above the ground level, as shown in Fig. 7. We empirically evaluate the node's coverage and find that it has an optimal detection area of 9 m^2. The monitored (squared) area consists of nine grids, of which has a size of 1 m $\times$ 1 m. We also divide each grid into four smaller cells that have the same area size of 0.5 m $\times$ 0.5 m. We record the ground truth using a video camera using 60 frames/s, and the PIR sensors' signals are synchronized with each recorded frame using a local desktop time.

B. Training and Test Data Collection

We collect a training data set by instructing a person to walk within the monitored area in different directions. Specifically, the person needs to walk forward and backward horizontally and vertically from six starting locations, thus resulting in 24 directions. Besides, the person needs to walk forth and back diagonally from the top left/right corner to the bottom right/left corner of the monitored area. In this case, there are 11 starting locations. Based on the above instruction, we record 44 directions, and we have 1020 sequences of PIR signals as our training data in total. To address the assumption of the constant walking speed, a metronome is employed to generate a steady pulse, which is set according to the person's walking speed preference. As a result, the person can use the pulse as a guideline to control a constant walking speed. Each collected PIR signal sequence is labeled using ground-truth coordinates extracted from the video camera. Our CNN-LSTM model is trained using MATLAB 2020b on i5 3.2-GHz Intel quad-core CPU, RAM 16 GB, and a GPU of NVIDIA RTX2070 with 8-GB video memory. We vary the model's hyperparameters to achieve the optimal results, and they are shown in Table I.

For performance evaluation, we use mean distance error and the cumulative distance function (CDF) of distance errors as a metric to measure the accuracy of our CNN-LSTM model. We design four walking scenarios that range from simple

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TABLE I
CNN-LSTM TRAINING PARAMETER SETTING

Parameter	Value
Max epoch	100
Mini batch size	32
Learning rate	0.0005
Learning rate Drop factor	0.1
Gradient threshold	1
Gradient Threshold Method	l2norm
L2 Regularization	0.0001

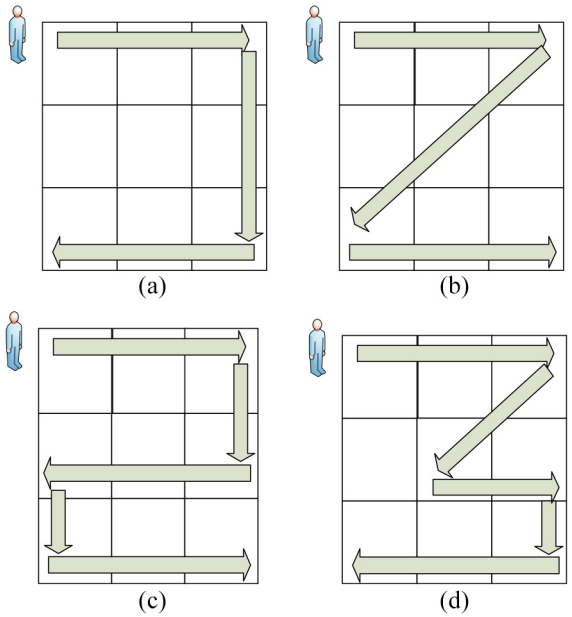


Fig. 8. Illustration of the walking scenarios. (a) Scenario 1. (b) Scenario 2. (c) Scenario 3. (d) Scenario 4.

walking patterns to more complex ones, as shown in Fig. 8. In the complex scenarios, we include a few short walking paths to test how well our model handles sharp turns and random walking directions. In addition to the location estimation accuracy, we are interested in the other factors that can affect the performance, including the impact of a sequence length in each observation and the impact of limited training data. We also compare different RNN models in our model, including LSTM, bi-LSTM, and GRU to compare the performance of each recurrent network. Besides the deep learning models, we select SVR in the comparison because it can provide accurate results compared with the other traditional methods and can even outperform deep learning in some cases [21], [62]. In this article, a linear kernel for SVR is used because it performs better than the other kernels in our cases. In addition, we compare our proposed method with PF which is a commonly used method for location estimation. We implement a system model and observation model based on the azimuth change measurement proposed in [15] for a PF method. In addition, we set the number of particles to 2000 particles because we have varied the number of particles and found that 2000 particles are the optimal one. Finally, a systematic method is employed to resample particles due to its simplicity.

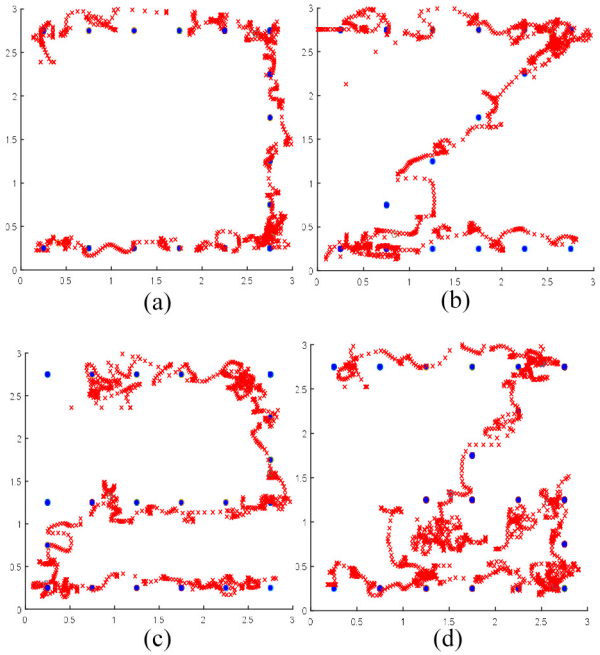


Fig. 9. PIR localization and tracking results. Red X represent estimated location at each time step. Blue dots represent the ground truth location. (a) Scenario 1. (b) Scenario 2. (c) Scenario 3. (d) Scenario 4.

TABLE II
MEAN DISTANCE ERRORS AND THEIR STANDARD DEVIATIONS FOR DIFFERENT SCENARIOS

SCENARIO	2 consecutive time steps		single time step	
	MEAN	Standard Deviation	Mean	Standard Deviation
1	0.1348 m	0.1202 m	0.1603 m	0.1159 m
2	0.3373 m	0.2050 m	0.5223 m	0.3497 m
3	0.1772 m	0.1450 m	0.2092 m	0.1613 m
4	0.3016 m	0.1968 m	0.3552 m	0.1379 m

TABLE III
MEAN DISTANCE ERRORS AND THEIR STANDARD DEVIATIONS FOR DIFFERENT MODELS AND TECHNIQUES

STRUCTURE	MEAN	Standard Deviation	Training Time	Estimation Time
CNN-LSTM	0.2379 m	0.1876 m	6.10 mins	0.043 secs
CNN-BiLSTM	0.3222 m	0.2316 m	10.12 mins	0.063 secs
CNN-GRU	0.5172 m	0.1924 m	5.34 mins	0.037 secs
SVR	0.6906 m	0.4303 m	0.23 mins	0.003 secs
Particle filter	0.4588 m	0.2718 m	N/A	0.320 secs

VI. RESULTS AND DISCUSSION

A. Location Estimation Accuracy

The accuracy of location estimation can be determined based on the Euclidean distance error between the ground truth's coordinates and estimated coordinates. Our CNN-LSTM model can achieve the overall mean distance error and the standard deviation of 0.2379 and 0.1876 m, respectively, as shown in Table III. In addition, our results show that 80% of the distance errors are less than 0.4 m, as shown in Fig. 11, which proves the feasibility of our approach for PIR-based location estimation. We can see that the estimated locations (red X) as shown in Fig. 9 follow the actual location (blue

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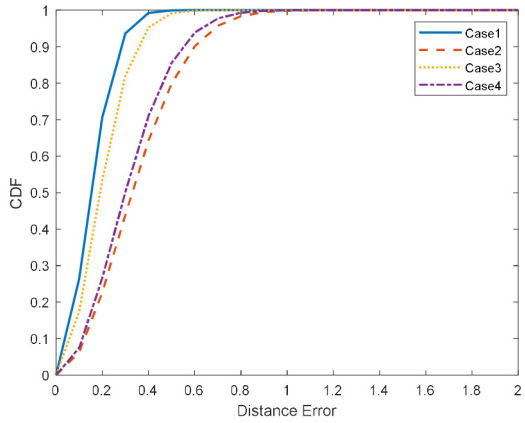


Fig. 10. CDF of distance error for each scenario.

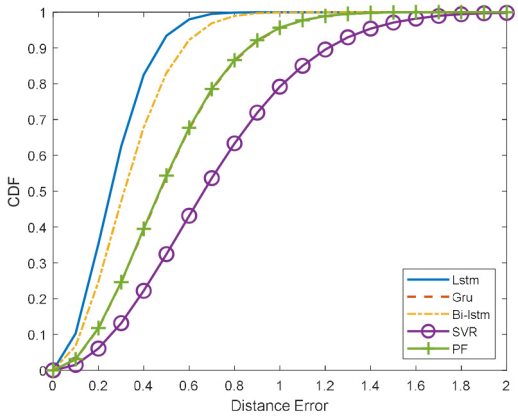


Fig. 11. CDF of distance error for each model structure (all scenarios).

dot) closely. For each scenario, the results are shown in column 2 of Table II. Scenario 1 achieves the mean distance error of 0.1348 m and the standard deviation of 0.1202 m. The CDF plot of Scenario 1 in Fig. 10 shows that 80% of the distance errors are less than 0.3 m. In the second scenario, we achieve the mean distance error of 0.3373 m and the standard deviation of 0.2050 m. The CDF of the second scenario shows that 70% of the errors are less than 0.4 m. In the third scenario, the mean distance error of 0.1772 m and the standard deviation of 0.1450 m is obtained. 70% of the distance errors are less than 0.3 m as shown in Fig. 10. The fourth scenario can achieve the mean distance error of 0.3016 m and the standard deviation of 0.1968 m. In addition, 70% of the mean distance errors are less than 0.4 m. From these four scenarios, our model is proven capable of localizing and tracking a person without the knowledge of the initial location. Although estimation is not accurate at the beginning, the more information is received, the better the model corrects itself and makes a better estimation. For each time step, the received signal data may be ambiguous in some cases, such as when a person starts moving or when a person walks under the PIR sensor node. We observe the amplitudes of raw signals are quite low, and the signal patterns seem to be significantly varied in comparison to the raw signal generated by a person walking perpendicular to the sensor node. Using the data from a single time step, the length of the data is not enough to provide any meaningful insight. As a result, the information stored in an updated hidden state, and an updated cell state is too ambiguous to make an accurate location estimation. This problem affects the accuracy of location estimation in every walking scenario, especially Scenario 2 that we obtain the highest mean distance error of 0.5223 m as shown in column 3 of Table II. Once we update the states of the LSTM layer using data from two consecutive time steps, we obtain the improved result for every scenario as shown in Table II. This result indicates that the information stored in the hidden and cell states is less ambiguous and can capture a relationship between the signal data for each time step. As a result, the locations of a person can be estimated accurately.

Next, we compare our model with the other well-known recurrent networks, including Bi-LSTM, GRU, SVR, and PF. Fig. 11 and Table III show that our model can achieve the best performance for both mean distance error and CDF. Our CNN-LSTM is about 30% better than CNN-Bi-LSTM. Although Bi-LSTM has an advantage over LSTM by taking into account both past and future information, this is not for our case because each sensor observation is received in real time so no future information available for BI-LSTM. The performance of CNN-GRU is about 73% lower than ours. GRU aims to improve training speed by simplifying the cell design to reduce the number of parameters, and the input and forget gates are combined in an update gate. In addition, GRU controls the flow of information without using a cell memory unit. However, it becomes less expressive compared with LSTM, and we find that GRU is not suitable for modeling correlations of PIR output sequences, and its results are inferior to LSTM in our cases. SVR can achieve promising results with the distance error of 0.69 m in our experiments, but its accuracy is still lower than the deep learning models used in our study. Finally, PF's accuracy is 63% lower than our CNN-LSTM. We observe that azimuth change measurements can be similar in several positions in the monitored area. Thus, distance errors increase considerably in several time steps due to the ambiguity of the measurement. In addition, the azimuth change requires extensive efforts to extract and enhance to make it usable, whereas deep learning methods can automatically extract and learn discriminative features. Thus, our proposed deep learning method is more practical.

B. Computational Time

Computational time is one important factor for a localization and tracking system because it can affect both the offline and online training phases. In the offline training, a long training time can delay the startup of the system. During the online phase, a long computational time can cause a delay in a location estimation, thus making results inaccurate and less useful. Our proposed model takes roughly 6 min to train for 50 epochs, as shown in Table III. The training duration is 50% faster than BiLSTM but it is 13% longer than the GRU model. However, our model is still favorable due to its better accuracy. For an estimation time, our model takes the average of only 0.043 s to estimate a location of a person at each time step, as shown in Table III. The CNN-GRU and

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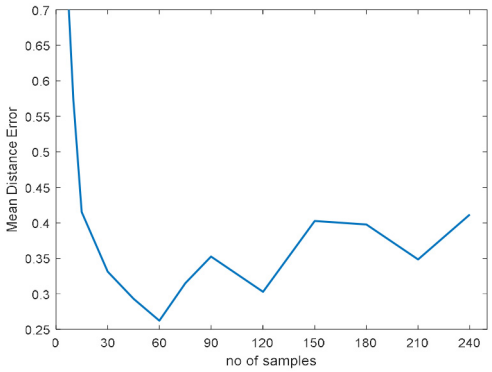


Fig. 12. Impact of number of samples in each observation on accuracy (all scenarios).

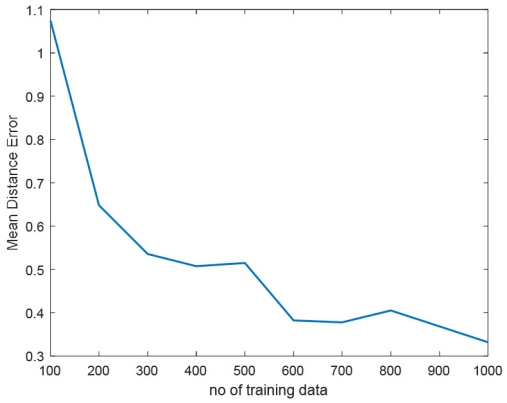


Fig. 13. Impact of size of training data on accuracy (all scenarios).

CNN-BiLSTM take 0.037 and 0.063 s, respectively. There is no significant difference in estimation times among these three models. SVR requires 0.23 min and 0.003 s to train and to estimate a location, respectively. Compared with the other methods, SVR is the fastest method in our comparison, but the difference is insignificant. The PF obtains the highest execution time because we need to perform calculations for all particles. Even so, the execution time of the particle filter is acceptable for real-time location estimation.

C. Impact of the Number of Samples in Each Observation

The number of samples in each observation as well as how often each observation is received can affect the accuracy of location estimation. A longer sequence may improve accuracy, but it takes a longer time to be received. Thus, it causes a delay in location estimation. Therefore, we try to evaluate our approach with the different number of samples in each observation, as shown in Fig. 12, to see how it affects the estimation accuracy. We found that the accuracy continues to improve with the increase in the number of samples in each observation. Scaling from 10 to 60 samples, the mean distance errors are decreased by 75%. However, the accuracy of our model is gradually increased when a sequence length increases above 60 samples. As our model utilizes two consecutive time steps to update the states of LSTM layers, when the length of data is too long, some irrelevant data are included in the hidden and cell states. As a result, these irrelevant data can affect the accuracy of the model. Based on these results, we can say that the selection of 60 samples in a sequence is optimal.

D. Impact of the Amount of Training Data

One main issue that keeps the deep learning methods away from deployment in real life is the limitation of training data. Constructing a large data set requires considerable manpower, which is time consuming and is not practical in real life. Therefore, we evaluate our approach with different sizes of training data, as shown in Fig. 13. The mean distance error of our model improves significantly with a training data size of 200. In addition, the distance error decreases steadily with the increase in the number of training data without the drastic rise and drop in the distance error. Therefore, we can use small and limited training data to initialize our model and gradually add new data to retrain the model. This result can prove that our model is feasible in a real-life environment.

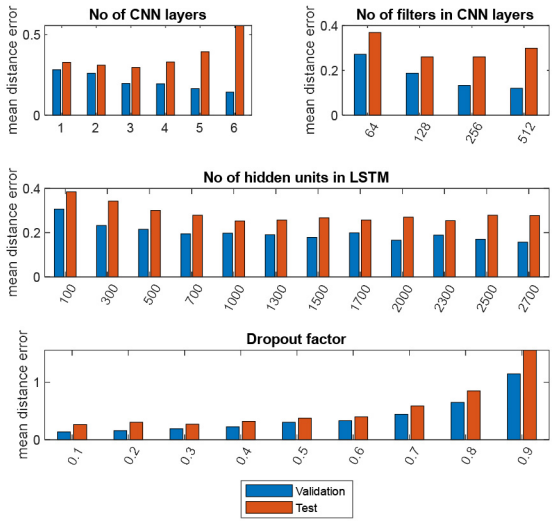


Fig. 14. Hyperparameter testing of CNN-LSTM model.

E. Model Hyperparameters

To discover the optimal hyperparameters of the proposed CNN-LSTM model, we conduct several experiments on the hyperparameters extensively. We test notable hyperparameters that affect the accuracy of the location estimation. The results of the hyperparameter experiments are illustrated in Fig. 14. For the number of CNN layers, we vary the number of CNN layers from one to six layers. An increment of the number of CNN layers can improve the accuracy of the location estimation using the validation data. However, the results of the test scenarios show otherwise. More layers can reduce the accuracy considerably due to overfitting. The optimal number of CNN layers is 3 in our case. The number of filters can affect the accuracy of the proposed model. We vary a set of filters, including 64–126–256, 128–256–512, 256–512–1024, and 512–1024–2048. A higher number of filters can improve the accuracy of the proposed CNN-LSTM model using the validation data, but the model can perform poorly in the test scenario because of the overfitting issue. The optimal number of filters is 128–256–512 in this case. The results of LSTM

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hidden units show the same trend for both validation and test. The higher the number of hidden units, the better the accuracy. We use 1024 units because the accuracy of the location estimation does not change significantly beyond this point. Finally, we vary a dropout factor from 0.1 to 0.9. The results show that the higher dropout value can reduce the accuracy of the location estimation because the model becomes too generalized. We set the drop factor to 0.3 because it gives the best results in our experiments.

VII. Conclusion and Future Works

In this article, we propose a deep CNN-LSTM architecture for indoor location estimation using PIR analog signals. Based on an assumption of a constant walking speed, our proposed model not only extract discriminate features from PIR signals without the need for handcrafted features but also capture the temporal pattern of the extracted features to accurately and timely estimate the location of a person. A set of experiments mimicking real-world walking scenarios are conducted to evaluate our model, and the results show that we can achieve a mean distance error of 0.23 m, and 80% of distance errors are within 0.4 m.

The presence of multiple persons in the monitored area can generate ambiguous signal patterns. As a result, it affects the accuracy of our proposed method. To handle multiple persons, some challenges need to be addressed, such as signal separation and data association. In this article, we scope our research problems to only a single person and will extend our approach to support multiple persons in the future.

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